

Multicenter randomized controlled trial of neoadjuvant chemoradiotherapy alone or in combination with pembrolizumab in patients with resectable or borderline resectable pancreatic adenocarcinoma

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ABSTRACT

Background Pancreatic ductal adenocarcinoma (PDAC) is a challenging target for immunotherapy because it has an immunosuppressive tumor microenvironment. Neoadjuvant chemoradiotherapy can increase tumor-infiltrating lymphocyte (TIL) density, which may predict overall survival (OS). We hypothesized that adding programmed cell death protein 1 (PD-1) blockade to chemoradiotherapy would be well tolerated and increase TILs among patients with localized PDAC.

Methods Patients were randomized 2:1 to Arm A (receiving pembrolizumab plus chemoradiotherapy (capecitabine and external beam radiation)) or Arm B (receiving chemoradiotherapy alone) before anticipated pancreatectomy. Primary endpoints were (1) incidence and severity of adverse events during neoadjuvant therapy and (2) density of TILs in resected tumor specimens. TIL density was assessed using multiplexed immunofluorescence histologic examination.

Results Thirty-seven patients were randomized to Arms A (n=24) and B (n=13). Grade ≥3 adverse events related to neoadjuvant treatment were experienced by 9 (38%) and 4 (31%) patients in Arms A and B, respectively, with one patient experiencing dose-limiting toxicity in Arm A. Seventeen (71%) and 7 (54%) patients in Arms A and B, respectively, underwent pancreatectomy. Median CD8⁺ T-cell densities in Arms A and B were 67.4 (IQR: 39.2–141.8) and 37.9 (IQR: 22.9–173.4) cells/mm², respectively. Arms showed no noticeable differences in density of CD8⁺Ki67⁺, CD4⁺, or CD4⁺FOXP3⁺ regulatory T cells; M1-like and M2-like macrophages; or granulocytes. Median OS durations were 27.8 (95% CI: 17.1 to NR) and 24.3 (95% CI: 12.6 to NR) months for Arms A and B, respectively.

Conclusions Adding pembrolizumab to neoadjuvant chemoradiotherapy was safe. However, no convincing effect on CD8⁺ TILs was observed.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Although many strategies have been studied to improve the response of pancreatic ductal adenocarcinoma (PDAC) to immunotherapy, none have translated to clinical benefit.

WHAT THIS STUDY ADDS

⇒ We hypothesized that chemoradiotherapy (CRT) could be used safely to prime the immune microenvironment and that a combination of CRT and immune checkpoint inhibition with a programmed cell death protein 1 (PD-1) inhibitor would increase tumor-infiltrating lymphocytes (TILs') ability to overcome the primary resistance of PDAC to PD-1 blockade. In this study, neoadjuvant PD-1 blockade concurrent with chemoradiotherapy was safe in the neoadjuvant setting. However, no salient effect on CD8⁺ TILs was observed.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ Future work regarding novel drug combinations should focus not only on priming adaptive immune responses but also targeting the PDAC immunosuppressive microenvironment.

INTRODUCTION

Over 60,000 people in the USA were diagnosed with pancreatic ductal adenocarcinoma (PDAC) in 2021, and only about 6,000 (10%) have a projected survival of 5 years or more.¹ Resection of pancreatic tumors and associated lymph nodes via pancreatectomy may cure PDAC in upwards of 20% of

patients; however, fewer than 20% of newly diagnosed patients have operable cancers, and most patients who do undergo surgery subsequently develop incurable, recurrent disease.² Neoadjuvant therapy is frequently administered to patients with resectable or borderline resectable PDAC to improve systemic disease control, increase the likelihood of a complete resection, and determine which patients will most likely benefit from surgery.³

Neoadjuvant therapy has historically consisted of chemotherapy, chemoradiotherapy (CRT), or both, but interest in the administration of immune checkpoint inhibitors in this setting has increased.^{4,5} Unfortunately, immune checkpoint inhibitors appear ineffective in most PDAC cases, as they have shown effectiveness in only the <1% of cases with microsatellite instability. This poor response may be due at least in part to the lack of activated intratumoral T cells and the complex immunosuppressive tumor microenvironment (TME) being dominated by myeloid-derived suppressor cells, regulatory T cells, and cancer-associated fibroblasts.^{6–12} Increased density of tumor-infiltrating lymphocytes (TILs) and tumor-neoantigen-specific memory CD8⁺ T cells has been associated with increased longevity of patients with PDAC, suggesting that priming the TME may improve the activity of immune checkpoint inhibitors.^{13–15}

CRT can enhance major histocompatibility complex (MHC) class I expression in tumor cells, thereby increasing their immunogenic properties. Tumor cells can then undergo immunogenic cell death that can release tumor antigens, activate local dendritic cells to migrate to the draining lymph nodes, and increase CD4⁺ and CD8⁺ T-cell priming.^{16–18} However, the ability of cancer cells to evade antitumor T-cell activity is controlled by immune checkpoint inhibitors such as programmed death-ligand 1 (PD-L1), which is expressed on both pancreatic cancer cells and immunosuppressive myeloid cells and is further induced by chemotherapies such as 5-fluorouracil and radiation therapy.^{19–21} Whether CRT can be combined with PD-1 inhibitors to increase cancer cell susceptibility to treatment remains unknown.

We hypothesized that CRT could be used safely to prime the immune microenvironment and that a combination of CRT and immune checkpoint inhibition (specifically, a PD-1 inhibitor) would increase TILs' ability to overcome the primary resistance of PDAC to PD-1 blockade. To test this hypothesis, we performed a prospective, multicenter, randomized controlled trial to assess the safety and preliminary estimates of the efficacy of treatment with neoadjuvant CRT with pembrolizumab compared with neoadjuvant CRT alone in patients with resectable or borderline resectable PDAC.

PATIENTS AND METHODS

Study design

This was a prospective, open-label, Phase Ib/II randomized clinical trial (ClinicalTrials.gov identifier NCT02305186) with 2:1 patient allocation to the treatment (Arm A) and

control (Arm B) arms. Arm A consisted of treatment with a combination of pembrolizumab and CRT (the 5-fluorouracil-based drug capecitabine and external-beam radiotherapy), and Arm B consisted of treatment with CRT alone. The primary endpoints were (1) the incidence and severity of adverse events (AEs) during neoadjuvant therapy and (2) the relative density of CD8⁺ TILs in resected tumor specimens. Secondary endpoints were treatment effect, progression-free survival (PFS), recurrence-free survival (RFS), and overall survival (OS).

The study was conducted at six sites: University of Virginia, Dana-Farber Cancer Institute, The University of Texas MD Anderson Cancer Center, Mayo Clinic, Hartford HealthCare Cancer Center, and University of Miami.

Inclusion criteria

Inclusion criteria comprised the presence of resectable or borderline resectable PDAC as classified according to the Alliance for Clinical Trials in Oncology criteria, with measurable disease identified by the Response Evaluation Criteria in Solid Tumors (RECIST) V.1.1 guidelines and an Eastern Cooperative Oncology Group performance score of 0 or 1.²² Exclusion criteria comprised the presence of metastatic disease; administration of immunosuppressive therapy within 7 days of the first trial treatment; and prior surgery, chemotherapy, targeted small-molecule therapy, immunotherapy, or radiation therapy for pancreatic cancer.

Protocol therapy and follow-up

Patients in both arms received CRT consisting of concurrent capecitabine (825 mg/m² orally two times a day) and external-beam radiation (50.4 Gy in 28 fractions, 5 days per week), starting on day 1. Patients in Arm A also received 200 mg of intravenous pembrolizumab every 3 weeks on days 1, 22, and 43.

A restaging CT scan was performed 4–6 weeks after completion of the neoadjuvant regimen. Patients without significant radiographic evidence of local or distant disease progression were scheduled to undergo pancreatectomy within 2 weeks of the restaging scan. Following pancreatectomy, patients were treated with adjuvant therapy at the discretion of their treating oncologist. The study design is shown schematically in [figure 1](#). Patients were followed-up every 4 months after the end of treatment until they reached 24 months post-registration.

Clinical assessment

Radiologic response at each institution was evaluated using the RECIST V.1.1 guidelines.²³ Surgical specimen analysis was performed following recommendations of the College of American Pathologists.^{24,25} Treatment effect in surgical specimens was assessed by an experienced pathologist at each study site. Treatment effect was scored as 0 (complete response, no viable cancer cells), 1 (near-complete response, single cells or rare small groups of cancer cells), 2 (partial response, residual cancer with evident tumor regression but more than single cells

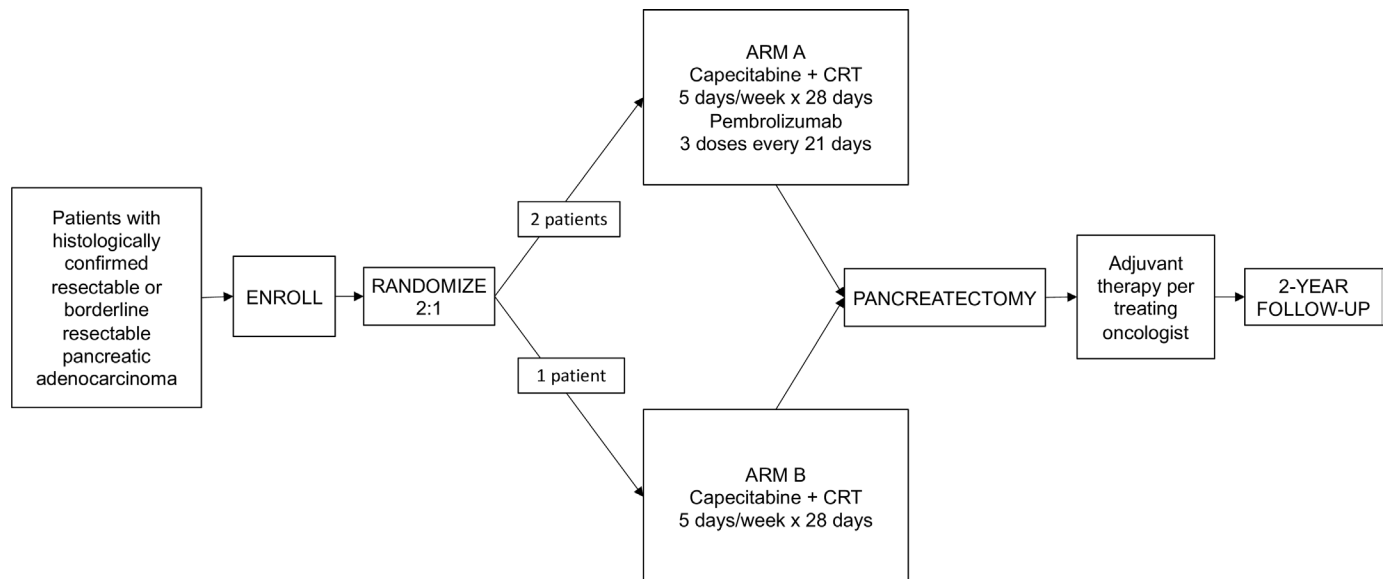


Figure 1 Schematic illustration of study design. Patients with histologically confirmed resectable or borderline resectable pancreatic ductal adenocarcinoma were enrolled and randomized in a 2:1 ratio to receive CRT plus pembrolizumab or CRT alone prior to anticipated pancreatectomy. During neoadjuvant therapy, peripheral blood mononuclear cells were collected for analysis by flow cytometry. At pancreatectomy, tissue was collected for analysis by immunofluorescence imaging. Patients were followed for 2 years while receiving adjuvant chemotherapy at the discretion of their treating oncologist. CRT, chemoradiotherapy.

or rare small groups of cancer cells), or 3 (poor or no response, extensive residual cancer with no evident tumor regression).

AEs were graded using the Common Terminology Criteria for Adverse Events V.4.0 and were monitored from the start of neoadjuvant treatment to 30 days following surgery or, among patients who did not undergo resection, to 30 days following the date of the restaging scan or anticipated surgery.²⁶ Participants treated on Arm A were also monitored for dose-limiting toxicities (DLTs) prior to surgery and for readmission within 30 days following surgery for events related to the combination.

Assessment of the tumor immune microenvironment

Formalin-fixed paraffin-embedded (FFPE) tissue from surgical resection specimens with residual tumor was analyzed using multiplexed immunofluorescence (mIF), digital image analysis, and machine learning to identify immune cell subsets in the TME. Three mIF antibody panels optimized for PDAC were used for analysis of T cells (CD4, CD8, FOXP3, Ki67), myeloid cell subsets and ARG-1 immunosuppressive potential (CD15, CD14, ARG1, HLA-DR, CD33), and macrophage polarization status (IRF5, CD163, CD68, CD86, MRC1 (CD206)); suppliers and dilution are provided in online supplemental table S1. All panels included 2-(4-amidinophenyl)-1H-indole-6-carboxamide (DAPI) as a nuclear marker and anti-cytokeratin to identify epithelial cells. Methodologic details for the myeloid cell subset and macrophage polarization panels have been previously published, while the T-cell panel was specifically built for this study using previously validated antibodies.^{27, 28} In brief, staining was performed using Leica BOND RX Research Stainer

(Leica Biosystems, Buffalo, Illinois, USA). Slides stained for the T-cell panel were digitized using a Mantra 2 multispectral imaging system (Akoya Biosciences, Hopkinton, Massachusetts, USA), while myeloid cell subset and macrophage polarization slides were digitized using a Vectra V.3.0 multispectral imaging system (Akoya Biosciences). For each slide, an average of 4 (range: 2–10) representative regions of interest containing residual, morphologically viable tumor epithelium were scanned at 200× magnification. Multispectral images were unmixed and processed to generate single-cell phenotypes using the *inForm* software package (Akoya Biosciences). The resultant single-cell level data were further analyzed using R V.4.0 (R Foundation for Statistical Computing, Vienna, Austria) to generate cell density measurements. Region-level density measurements from each panel were averaged to produce patient-level immune cell density measurements.

Immunohistochemical analysis of PD-L1 was conducted on FFPE specimens using a monoclonal anti-PD-L1 antibody (clone E1L3N, Cell Signaling Technology, Danvers, Massachusetts, USA, dilution 1:100). PD-L1 expression was assessed by determining the Tumor Proportion Score (TPS) and Combined Positive Score (CPS) across all viable tumor-containing regions of each slide.

Flow cytometry

Flow cytometric analyses of peripheral blood mononuclear cells (PBMCs) were conducted during neoadjuvant therapy to study the impact of CRT with and without pembrolizumab on peripheral immune cell populations. PBMCs were thawed at 37°C and transferred into 15 mL Falcon tubes containing 10 mL of Roswell Park

Memorial Institute (RPMI) medium (Gibco) supplemented with 10% fetal bovine serum (FBS, Gibco) and DNase I (2 U/mL, Sigma-Aldrich). These tubes were centrifuged at 350×g for 5 min at 4°C. Cell pellets were resuspended in 1 mL of fluorescence-activated cell-sorting (FACS) buffer (phosphate-buffered saline with 10% FBS and 2 mM EDTA). Each cell suspension was transferred into two Eppendorf DNA LoBind tubes, which were centrifuged at 350×g for 5 min at 4°C. The supernatant was discarded, and cell pellets were resuspended with 100 µL of FACS buffer and 2.5 µL of each of the following antibodies all from BioLegend: CD45RO BV650, HLADR BV570 clone L243, CD38 BV711, CD16 BV785 clone 3G8, CD14 BV605, CD45RA FITC, CX3CR1 BV421, KLRG1 PerCP-Cy5.5, CD19 AF700, CCR7 PE-Cy7, PD-1 PE, CD8 PB. After staining for 20 min at 4°C, 1 mL of FACS buffer was added to each tube, and the tubes were again centrifuged at 350×g for 5 min at 4°C. The supernatant was discarded, and the samples were fixed by adding 100 µL of 1% formalin. Flow cytometry analysis was performed using a SONY SP6800 Spectral Cell Analyzer (Sony Biotechnology, San Jose, California, USA).

Statistical considerations and analysis

Randomization and sample size calculation

Participants were stratified by resectable or borderline resectable disease. Participants were then randomized in a 2:1 ratio to Arms A and B, respectively, within each stratum using a randomized block design. The intention-to-treat population was all randomized participants. The evaluable population was defined as participants who underwent pancreatectomy and had tissue available for TIL analysis. The overall target sample size of evaluable participants was 45, which was based on estimating the potential difference in the density of TILs in resected tissue between participants receiving neoadjuvant CRT in combination with pembrolizumab and participants receiving neoadjuvant CRT alone. This study was designed to obtain preliminary data on the combination and to estimate the magnitude of potential differences in immune cell populations for further research, not to directly compare endpoints between arms.

In a non-randomized study of patients with PDAC who underwent surgical resection with or without neoadjuvant CRT, the reported mean (±SD) numbers of CD8⁺ TILs per high-powered field (hpf) were 50.1 (±38.7) in patients who underwent pancreatectomy for PDAC after receiving CRT and 24.1 (±21.3) in patients who underwent pancreatectomy for PDAC and had not received prior therapy.¹⁸ Thus, using 38.7 as a conservative estimate of the SD of the difference in CD8⁺ TILs before and after resection, we determined that evaluable sample sizes of 30 and 15 for Arms A and B, respectively, would produce a two-sided 95% CI for a difference of 25 TILs per hpf between arm means.

Endpoint analyses

The first primary endpoint was the incidence and severity of AEs during neoadjuvant therapy. A DLT was defined as a grade 3 or higher non-hematological or grade 4 hematological AE considered to be related to the combination of pembrolizumab and CRT or a grade 3 or higher infusion reaction due to pembrolizumab. To monitor for excessive DLT and readmission rates, the upper boundary of a sequential probability ratio test (SPRT) based on a binomial test of proportions with nominal type I and II errors of 10% each was defined separately. Results from previous studies using only neoadjuvant CRT in PDAC indicated an unacceptable rate of AEs of approximately 5% and provided the basis for an SPRT contrasting a 5% versus 25% DLT rate.²⁹ A readmission rate following pancreaticoduodenectomy in the USA of 21.3% (95% CI: 19% to 24%) provided the basis for an SPRT contrasting a 20% and 36% 30-day readmission rate.³⁰

The second primary endpoint was the relative density of CD8⁺ TILs in resected tumor specimens. Estimated mean differences and 95% CIs between immunologic effects before and after treatment between arms assumed a pooled variance and a t-distribution. Given that only 53% of the target sample size was achieved and that the observed variability in CD8⁺ TILs was larger and more skewed than was assumed, quantile regression with no intercept was used to estimate median values and differences in medians with 95% CIs of immunologic effects between arms.

Secondary endpoints were treatment effect, PFS (defined as the time from start of treatment until first documentation of progression, recurrence, or death or until the date of last follow-up for those who had not had progression, recurrence, or death), RFS (defined as the time from pancreatectomy until first documentation of recurrence or death or until the date of last follow-up for those who had not had recurrence or died), and OS (defined as the time from start of treatment until death from any cause or until the date of last follow-up for those still alive). For each arm, time-to-event distributions were estimated by the Kaplan-Meier product limit method. We also calculated estimates and 95% CIs for median and yearly survival.

Data availability

The data generated in this study are available on request from the corresponding author.

RESULTS

Study population

The study was designed to accrue up to 68 patients to obtain 45 evaluable participants. Between August 2015 and November 2019, 37 patients were enrolled and randomized to Arm A (n=24) and Arm B (n=13). Due to slow enrollment and perceived changes in standard of care,^{22 31} the protocol was amended on December 31, 2019, to allow inclusion of patients who had previously

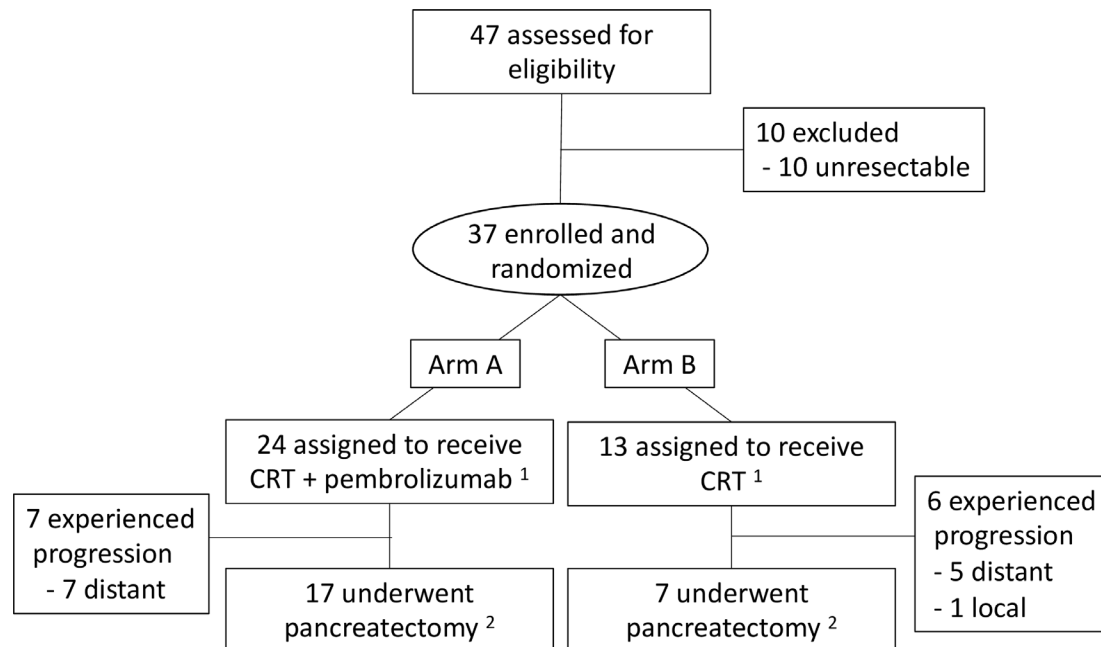


Figure 2 Consolidated Standards of Reporting Trials diagram of patient groups. ¹ Evaluable for safety endpoint (n=37). ² Evaluable for immunologic endpoint (n=24). CRT, chemoradiotherapy

received FOLFIRINOX, but the study was terminated because of slow accrual after the enrollment of five additional patients.

This report describes the 37 patients enrolled prior to the amendment, who had not received prior therapy. The Consolidated Standards of Reporting Trials diagram is presented in [figure 2](#). The baseline demographic and clinical characteristics of these 37 participants are reported in [table 1](#).

Neoadjuvant therapy

All 37 patients initiated CRT per-protocol. During CRT, 19 (79%) patients on Arm A and 11 (85%) patients on Arm B completed the prescribed dose of radiation.

Among patients on Arm A, 19 (79%) patients received all three pembrolizumab doses per-protocol. Among the remaining 5 (21%) patients on Arm A, 4 (17%) patients received two doses, and 1 (4%) patient received one dose.

Adverse events

At least one grade 3 or higher AE considered at least possibly related to neoadjuvant treatment occurred in 9 (38%) patients in Arm A and 4 (31%) patients in Arm B ([table 2](#)). The only grade 4 AE at least possibly related to treatment was leukemia, experienced by one patient in Arm A; it was attributed to chemotherapy. Only one patient in Arm A experienced an AE that was deemed probably related to the combination of pembrolizumab and CRT and that qualified as a DLT (grade 3 elevation of alanine aminotransferase (transaminitis)). This AE resolved after withholding treatment and providing steroid therapy.

Surgery

Of the 37 patients in our study, 24 patients across Arm A (n=17) and Arm B (n=7) underwent surgery and were evaluable. Following neoadjuvant therapy, 17 (71%) patients in Arm A underwent pancreatectomy while 7 (29%) patients did not due to the interval development of metastasis. In Arm B, 7 (54%) patients underwent pancreatectomy while 6 (46%) patients did not due to the interval development of metastasis (n=5, 38%) or local disease progression (n=1, 8%).

Following surgery, 1 (6%) patient of the 17 patients who underwent pancreatectomy on Arm A was readmitted within 30 days of the procedure for reasons unrelated to the neoadjuvant treatment. Adjuvant chemotherapy was initiated in 16 (94%) resected patients in Arm A and 7 (100%) resected patients in Arm B. Surgical outcomes and pathologic staging information are summarized in online supplemental table S1. Fourteen (82%) resected patients on Arm A and 4 (57%) resected patients on Arm B had microscopically negative (R0) margins.

Survival

The median PFS durations were 18.2 (95% CI: 9.4 to 27.0) months for Arm A and 14.1 (95% CI: 2.6 to 24.3) months for Arm B ([figure 3A](#)). The median OS durations were 27.8 (95% CI: 17.1 to NR) months for Arm A and 24.3 (95% CI: 12.6 to NR) months for Arm B ([figure 3B](#)). Among patients who underwent pancreatectomy, the median RFS durations were 22.3 (95% CI: 13.5 to NR) months for Arm A and 21.8 (95% CI: 9.4 to NR) months for Arm B ([figure 3C](#)).

Table 1 Baseline demographics and staging information for all randomized patients (n=37)

Characteristic	Arm A: CRT+pembrolizumab (n=24)	Arm B: CRT (n=13)	Total (n=37)
Age, median years (range)	66 (53–85)	64 (57–79)	66 (53–85)
Race, n (%)			
White	22 (92)	12 (92)	34 (92)
Black or African American	1 (4)	0	1 (3)
Other	1 (4)	0	1 (3)
Unknown	0	1 (8)	1 (3)
Ethnicity, n (%)			
Non-Hispanic	20 (83)	11 (85)	31 (84)
Hispanic or Latino	3 (13)	0	3 (8)
Unknown	1 (4)	2 (15)	3 (8)
Sex, n (%)			
Male	13 (54)	6 (46)	19 (51)
Female	11 (46)	7 (54)	18 (49)
Clinical tumor stage, n (%)			
Resectable	10 (42)	6 (46)	16 (43)
Borderline resectable	14 (58)	7 (54)	21 (57)
Tumor location, n (%)			
Head/uncinate	17 (71)	10 (77)	27 (73)
Body	4 (17)	3 (23)	7 (19)
Body and tail	2 (8)	0	2 (5)
Head and body	1 (4)	0	1 (3)
TNM tumor stage, n (%)			
IA	3 (13)	1 (8)	4 (11)
IB	5 (21)	7 (54)	12 (32)
IIA	7 (29)	1 (8)	8 (22)
IIB	5 (21)	1 (8)	6 (16)
III	4 (17)	3 (23)	7 (19)
CA19-9			
Median (range)	108 (2–5,176)	71 (15–675)	106 (2–5,176)
Not elevated, n (%)	8 (33)	2 (15)	10 (27)
>ULN, n (%)	16 (67)	11 (85)	27 (73)
ECOG PS, n (%)			
0 (fully active)	13 (54)	7 (54)	20 (54)
1 (restricted)	11 (46)	6 (46)	17 (46)

CRT, chemoradiation; ECOG PS, Eastern Cooperative Oncology Group performance score; TNM, tumor-nodes-metastasis; ULN, Upper Limit of Normal.

Immune cell infiltration in the tumor microenvironment

The median CD8⁺ T-cell densities were 67.4 (IQR: 39.3–141.8) and 37.9 (IQR: 22.9–173.4) cells/mm² in Arms A and B, respectively (difference in medians: 29.5 (95% CI: –113.7 to 52.6); [table 3](#) and [figure 4](#)). The median CD4⁺ T-cell densities in resected tumor specimens were 119.3 (IQR: 46.1–156.3) and 48.4 (IQR: 40.7–223.8) cells/mm² in Arms A and B, respectively (difference in medians:

70.8 (95% CI: –63.9 to 205.6)). As expected, FOXP3 was expressed in a small subset of CD4⁺ cells, and densities of CD4⁺FOXP3⁺ regulatory T cells were similar between arms. Similarly, minor subsets of both CD4⁺ and CD8⁺ cells expressed the proliferation marker Ki67. Resected tumor specimens from patients in Arm A showed lower median densities of CD4⁺Ki67⁺ cells and higher median densities of CD8⁺Ki67⁺ cells compared with those for

Table 2 Grade 3 and 4 AEs observed during neoadjuvant treatment (N=37)

Adverse event	Arm A: CRT+pembrolizumab (n=24)		Arm B: CRT (n=13)	
	Grade 3	Grade 4	Grade 3	Grade 4
Gastrointestinal				
Diarrhea	2	0	0	0
Nausea	1	0	0	0
Hepatic				
Transaminitis	1*	0	0	0
Hematologic				
Lymphocyte count decreased	7	0	4	0
Leukemia	0	1†	0	0
Metabolic/nutrition				
Hypophosphatemia	1	0	0	0
Musculoskeletal				
Myalgia	1	0	0	0
Arthralgia	1	0	0	0

9 Patients had a grade 3 or 4 AE; some patients had more than one.
 *Attributed “probably” to CRT+pembrolizumab, thus qualifying as a dose-limiting toxicity.
 †Attributed to chemotherapy.
 AE, adverse event; CRT, chemoradiotherapy.

patients in Arm B. However, no differences in T-lymphocyte population density were observed between treatment arms, nor did the ratios of CD8⁺ to CD4⁺ T lymphocytes differ between treatment arms.

Neoadjuvant CRT has also been linked to changes in the myeloid cell landscape of pancreatic cancer.^{27,28} We therefore expanded our analysis to major monocyte/macrophage and granulocyte populations (table 3 and figure 4). Similar to the T-lymphocyte analyses, myeloid cell densities were similar in tumors for both study arms, although trends were noted. Overall, densities of CD15⁺ granulocytes and CD14⁺ monocytes were similar between treatment arms. In patients in Arm A, the median density of CD15⁺ARG1⁺ immunosuppressive granulocytes was approximately half that of Arm B. Finally, densities of total macrophages (CD68⁺) and/or M2-polarized macrophages (CD163⁺CD68⁺) were determined. Although the IQRs overlapped substantially, the median number of total macrophages per mm² was slightly higher in Arm A tumors, but the median number of M2-polarized macrophages per mm² was slightly lower in Arm A tumors. The study was not powered for comparison between arms, and the range of values overlapped substantially for each of these cellular subsets.

Beyond immune cell density, expression of PD-L1, the ligand for PD-1, could also be influenced by treatment with pembrolizumab. We therefore compared PD-L1 expression by immunohistochemistry between arms. As there is no consensus method for scoring PD-L1 expression in PDAC, we generated both TPS and CPS. Across

17 evaluable tumors, expression of PD-L1 was very low, with 88% of tumors showing a TPS of <1% or a CPS of <1 (online supplemental table S3). While the two tumors with the highest PD-L1 expression levels were in Arm A, both of these tumors harbored TPS and CPS values that are low on an absolute scale, and there was no statistically significant difference in TPS or CPS between arms.

Peripheral immune cell analyses

We performed flow cytometry to immunophenotype peripheral T cells over the course of neoadjuvant therapy. To verify on-target binding of pembrolizumab, we determined that PD-1 staining on CD8⁺CD45RO⁺ T cells was reduced in Arm A but not in Arm B, reflecting successful occlusion of the epitope by pembrolizumab. However, this effect was not uniform across the cohort, and three patients who did not derive clinical benefit showed an increase in PD-1⁺ CD8⁺ T cells (online supplemental figure S1A). In both arms, activated CD8⁺ T cells, marked by co-expression of HLA-DR and CD38,³² expanded during neoadjuvant therapy (mean of difference between end-of-treatment and baseline samples: 3.83 (95% CI: -0.02 to 7.68) in Arm A and 8.05 (95% CI: -0.88 to 16.98) in Arm B) (online supplemental figure S1B).

CX3CR1 has also been reported to be associated with PD-1 blockade response.³³ Among the patients in our study, the fraction of CD8⁺CD45RO⁺ T cells expressing CX3CR1 increased during neoadjuvant therapy (mean of difference between end-of-treatment and baseline samples: 9.44 (95% CI: 3.27 to 15.61)), although this

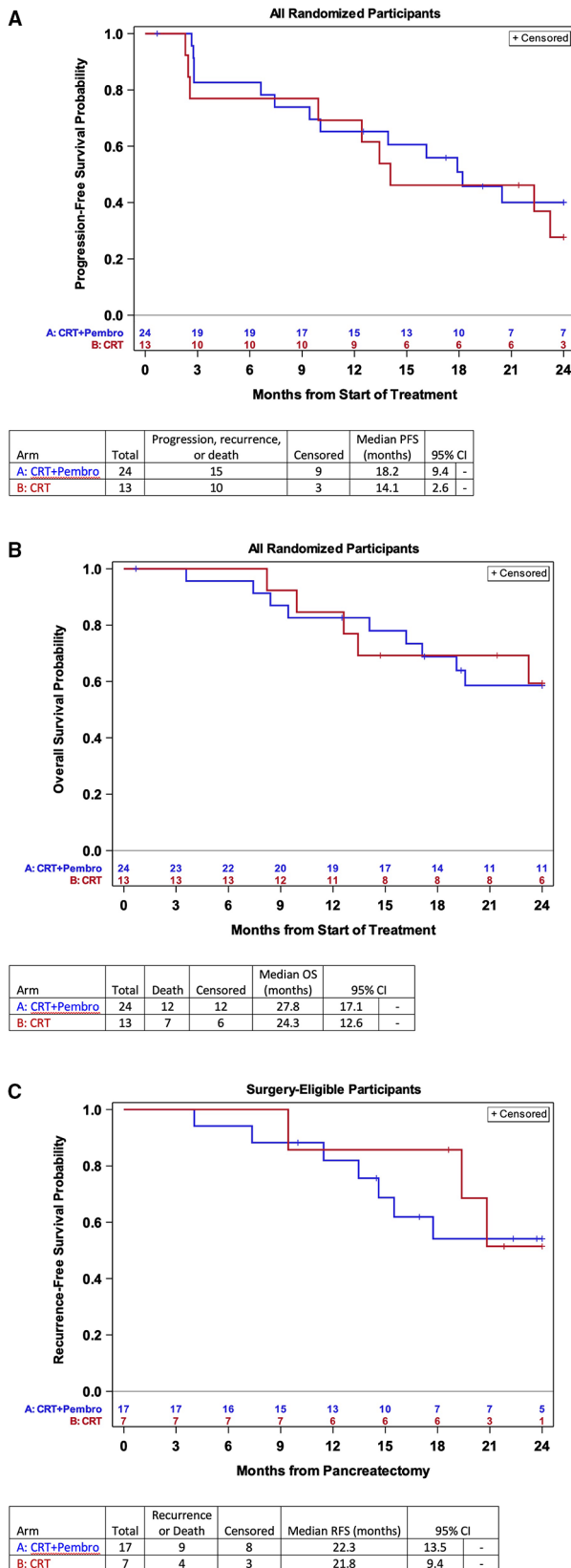


Figure 3 Survival analyses. Kaplan-Meier curves depicting (A) progression-free survival (PFS); (B) overall survival; and (C) recurrence-free survival of patients who underwent pancreatectomy. Arm A (CRT plus pembrolizumab) is depicted in blue while Arm B (CRT alone) is depicted in red. CRT, chemoradiotherapy.

increase was not seen when considering only Arm A (online supplemental figure S1C). Overall, we examined the density of HLA-DR, CD38, and CX3CR1—three flow cytometry-based biomarkers previously reported in the literature to describe checkpoint-blockade reactivated CD8⁺ T cells—but could not identify changes in these markers that were dependent on the addition of pembrolizumab.

DISCUSSION

This multicenter Phase Ib/II clinical trial was designed to test the safety and immunologic effect of the combination of a neoadjuvant PD-1 inhibitor, pembrolizumab, and CRT in patients with resectable to borderline resectable PDAC. The combination of CRT and pembrolizumab in the neoadjuvant setting was well tolerated by patients: only one DLT (grade 3 transaminitis) was observed, and there was no difference in postoperative complications between treatment with the combination and treatment with CRT alone. However, the addition of pembrolizumab to CRT did not have a detectable effect on the density of several immune cell populations, including CD8⁺ TILs, in the pancreatic cancer microenvironment. Furthermore, the median PFS, RFS, and OS durations of patients in both arms were similar.

Although many strategies have been studied to improve the response of PDAC to immunotherapy, none have translated to clinical benefit.³⁴ The failure of these studies to demonstrate a meaningful clinical benefit could be attributed to several factors: (1) inclusion of gemcitabine-based over 5-fluorouracil-based regimens due to its better safety profile, with little data on how these two regimens differ in their immunomodulatory effects; (2) testing immune modulators in the advanced metastatic setting, where there is limited time to generate a meaningful immune response in a patient with an exhausted immune repertoire; (3) testing concurrent rather than sequential administration of chemotherapy and immune modulators when chemotherapy may have a detrimental effect on the T cells needed to generate an antitumor immune response; or (4) lack of tumor-specific T cells available for reactivation by immune checkpoint inhibitors, potentially due to profound defects in naïve T-cell priming. Although we have not addressed all these potential causative factors here, we evaluated PD-1 blockade in combination with 5-fluorouracil-based CRT in the non-metastatic setting and found no clear benefit over treatment with 5-fluorouracil-based CRT alone. Still unanswered is whether the results of immunotherapy can be improved with sequential administration of CRT and PD-1 blockade in this setting, use of different radiation strategies, or targeting subsets of patients who have high primed T-cell density.

Notably, the density of CD8⁺ TILs was similar across tumors from both arms. Although some potential differences were observed in myeloid cell populations, these were neither substantial nor definitive. In circulating immune

Table 3 Means, medians, and 95% CIs for immune cell densities within whole tissue areas (combined intraepithelial and stromal areas) in patients treated with neoadjuvant chemoradiotherapy (CRT) plus pembrolizumab or neoadjuvant CRT alone

Tumor immune microenvironment (cells/mm ²)												
Immune cell subset	Arm A		Arm B		Arms A-B		Arm A		Arm B		Arms A-B	
	Mean (SD)	n=17	Mean (SD)	n=7	Difference mean	95% CI	Median (IQR)	n=17	Median (IQR)	n=7	Difference median	95% CI
T-cell panel												
CD8 ⁺	119.8 (140.5)		83.9 (97.1)		35.9	-70.1 to 142.0	67.4 (39.3-141.8)		37.9 (22.9-173.4)		29.5	-89.0 to 148.0
CD8 ⁺ Ki67 ⁺	17.1 (21.7)		21.6 (46.1)		-4.5	-47.3 to 38.4	7.6 (3.2-29.6)		2.8 (1.4-15.2)		4.8	-9.8 to 19.5
CD4 ⁺	112.5 (80)		124.3 (126.9)		-11.8	-130.9 to 107.4	119.3 (46.1-156.3)		48.4 (40.7-223.8)		70.8	-63.9 to 205.6
CD4 ⁺ FOXP3 ⁺	11.3 (15.8)		20.9 (28.4)		-9.6	-36.1 to 16.9	5.7 (2.1-13.5)		4.8 (2.1-38.7)		0.9	-23.5 to 25.3
CD4 ⁺ Ki67 ⁺	6.1 (6.5)		7.6 (8.2)		-1.5	-9.3 to 6.3	3.4 (1.9-8.0)		7.0 (0.7-14.6)		-3.6	-13.3 to 6.1
Myeloid-cell subsets		n=15		n=7				n=15		n=7		
CD15 ⁺ granulocytes	79.3 (136.8)		51.0 (63.2)		28.3	-60.6 to 117.3	14.9 (5.9-90.5)		17.2 (10.7-110.8)		-2.4	-78.3 to 73.5
CD15 ⁺ ARG1 ⁺	42.0 (67.0)		22.1 (26.7)		19.9	-21.9 to 61.7	4.2 (1.8-71.9)		9.0 (2.0-38.8)		-4.8	-44.6 to 35.0
CD15 ⁺ ARG1 ⁻	28.5 (55.6)		16.8 (18.8)		11.6	-21.9 to 45.1	11.9 (2.1-32.9)		8.7 (3.7-27.3)		3.1	-17.6 to 23.8
CD14 ⁺ monocytes	355.2 (245.7)		555.9 (530.3)		-200.7	-694.6 to 293.2	320.6 (129.1-539.1)		281.1 (189.1-767.1)		39.6	-393.1 to 472.2
Macrophages		n=12		n=7				n=12		n=7		
Total macrophages	1139.4 (1417.1)		1020.8 (879.5)		118.6	-994.2 to 1231.5	802.3 (482.3-943.4)		597.9 (511.3-1564.1)		204.4	-534.0 to 942.7
M1-polarized macrophages	365.9 (785.9)		133.5 (103.7)		232.4	-270.9 to 735.8	101.3 (52.4-180.0)		107.6 (44.6-205.9)		-6.3	-136.6 to 124.0
M2-polarized macrophages	213.0 (139.3)		343.2 (288.8)		-130.1	-400.5 to 140.2	199.5 (107.4-239.2)		325.6 (70.1-487.7)		-126.1	-407.7 to 155.5
CI, Confidence Interval; IQR, Interquartile range; SD, Standard Deviation.												

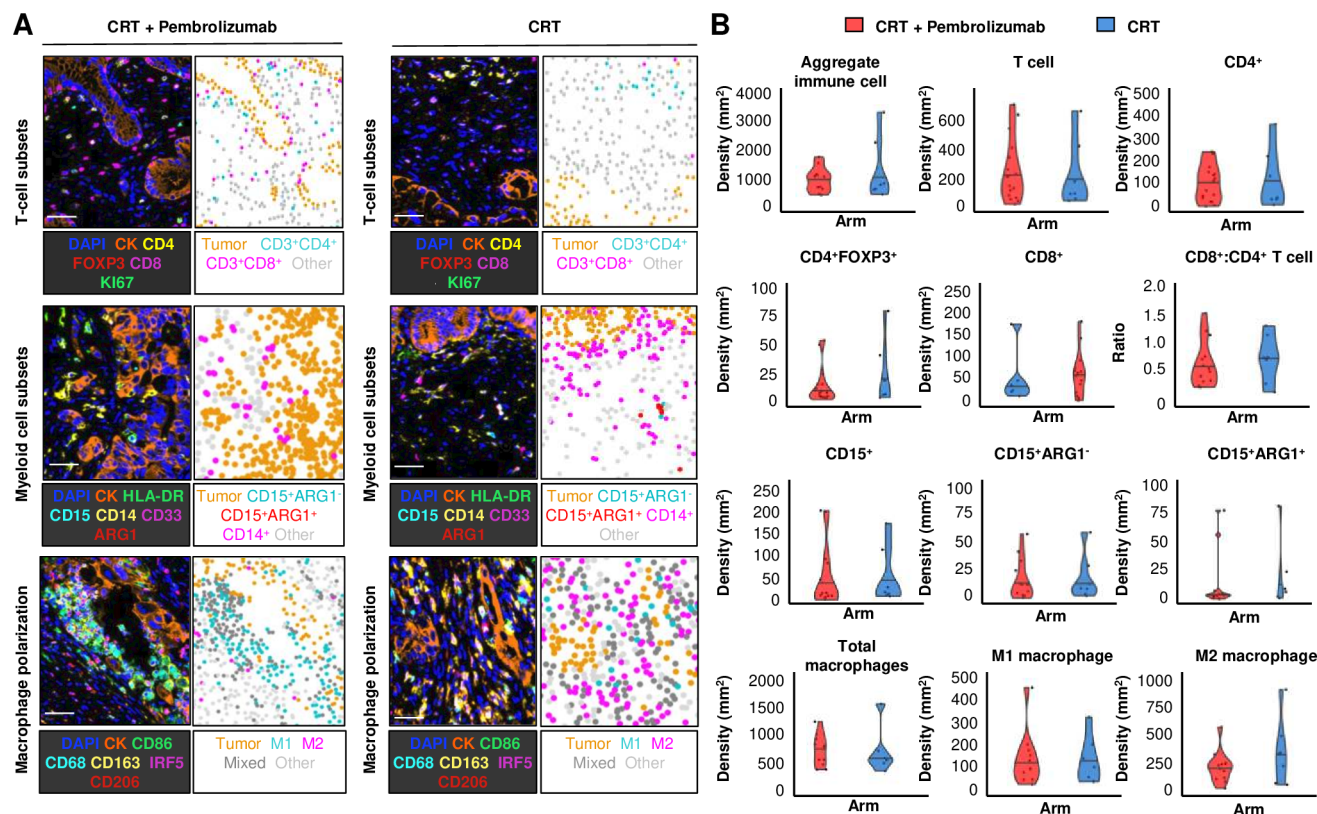


Figure 4 Immune landscape in resected pancreatic ductal adenocarcinoma (PDAC), stratified by treatment setting. Examples of multiplex immunofluorescence images and corresponding phenoplots for T-cell subsets, myeloid cell subsets and macrophage polarization panel in neoadjuvant chemoradiation (CRT) plus pembrolizumab-treated and neoadjuvant CRT-treated PDACs (A). Boxplots showing the distribution of overall (combined intraepithelial and stromal areas) immune cell densities in 15 neoadjuvant CRT plus pembrolizumab-treated and 7 neoadjuvant CRT-treated PDACs (B). Scale bars represent 50 μ m. DAPI, 2-(4-aminophenyl)-1H-indole-6-carboxamide.

cell populations, we found an increase in activation in a unique T-cell population (CD8⁺CD45RO⁺CX3CR1⁺ cells), suggesting that modest CD8⁺ T-cell activation can occur in the setting of CRT. However, the combination of pembrolizumab and CRT using capecitabine did not lead to statistically significant increases in T-cell activation in the blood or in the tumor, as measured by other standard markers and methodologies. It is possible that reactivated T cells in patients with PDAC display a different phenotype from reactivated T cells in patients with melanoma and other tumor types, or T cells in patients with PDAC express an exhaustion phenotype that cannot be reversed by the blockade of a single immune checkpoint.³⁵ Although pretreatment biopsies were not available for analysis of PD-L1 expression as a predictive biomarker in the study, the very low level of PD-L1 expression detected in treated tumors is consistent with other studies of PDAC and suggests that the PD-1/PD-L1 axis may not be highly operative in the PDAC immune microenvironment.³⁶ Immune checkpoint blockade may also be insufficient to change the PDAC TME to a favorable TME, and concurrent targeting immunosuppressive cells, including macrophages, may be needed.³⁷ However, the observed CD8⁺ TIL data in this study population was more variable and skewed than predicted, resulting in wider CIs

than originally assumed. Our findings, therefore, do not obviate the need for additional studies in this space.

Neoadjuvant treatment with FOLFIRINOX has been associated with a favorable immune profile characterized by increased CD8⁺ T lymphocytes, a shift in macrophage polarization toward an M1-like phenotype, and a reduction in immunosuppressive granulocytes in resectable PDAC.²⁸ These observations support our decision to amend the protocol to include patients who received prior FOLFIRINOX; however, we eventually had to close the study prematurely due to a slow accrual rate. The slow accrual rate can be explained, at least in part, by an evolving perception of the role of radiation in the neoadjuvant setting.^{38,39} Although studying the biological effects of immunotherapy in PDAC is appropriate in the preoperative period and we have demonstrated that such studies may be conducted effectively outside the National Clinical Trials Network cooperative group mechanism, nimbler study designs, such as adaptive platform studies, should be used in the future to rapidly test multiple agents and combinations for biological effect. Further, to the extent that patients in this study were disproportionately white, future studies should place greater effort on the enrollment of a diverse patient sample (online supplemental table S4).

Other limitations of this study pertain to our analysis of resected tumor specimens. For example, although our analysis of the TME used highly sensitive multiplexed immunofluorescence assays and targeted numerous broad classes of T cell and myeloid cells, these assays did not include markers for all potentially relevant immune cells and therefore could not capture all potential alterations that may occur due to treatment with a combination of CRT and pembrolizumab. In addition, although we profiled multiple regions from each tumor block to account for spatial heterogeneity, the total area of tissue analyzed per tumor was limited and may have influenced the accuracy of immune cell density estimates. Finally, while we appreciate the potential value of more comprehensive profiling of the patients and tumors in this study, the time period during which this trial was conducted predated widespread recognition of the value of somatic profiling and routine germline testing of patients with PDAC. Further, the limited tumor tissues available following clinical surgical pathology evaluation restricted our ability to perform next generation sequencing of tumor samples. In this regard, it must be noted that nearly all pancreatic adenocarcinomas have a low TMB^{40–43} and typically share highly recurrent alterations in a small number of driver genes.^{40 41 44} Indeed, in our prior studies of primary resected PDAC, across 300 tumors we did not find associations between the status of driver gene alterations (*KRAS*, *TP53*, *CDKN2A*, and *SMAD4*) or TMB and immune landscape as defined by density of more than a dozen separate lymphoid and myeloid immune cell populations.^{27 28 44} Given the high rates of alterations in common driver genes, and the low rate of DNA mismatch repair (MMR) deficiency and hypermutation in PDAC, we believe that it is unlikely that our results are significantly influenced by specific combinations of driver genes or tumors with an elevated TMB. Nevertheless, we certainly encourage future studies of immune targeting agents to include MMR status, in addition to genomic driver status and TMB.

In conclusion, the combination of CRT and pembrolizumab was found to be safe in the neoadjuvant setting in this randomized Phase Ib/II study. However, the addition of pembrolizumab to CRT did not have a detectable effect on several immune cell populations, including CD8⁺ TILs, in the pancreatic cancer microenvironment. Future work regarding novel drug combinations should focus not only on priming adaptive immune responses but also targeting the PDAC immunosuppressive microenvironment.

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